# Deficiency Manifestations Of Different Vitamins (Clinical scenarios)

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# 1. MATCH THE VITAMIN DEFICIENCY TO THE CLINICAL CONSEQUENCE.

**Q. 1 Bleeding**

**Q. 2 Diarrhea and dermatitis**

**Q. 3 Neural tube defects**

**Q. 4 Night blindness**

**Q. 5 Sore, spongy gums and loose teeth**

**A. Folic acid**

**E. Vitamin C**

**B. Niacin**

**F. Vitamin D**

**C. Vitamin A**

**G. Vitamin E**

**D. Vitamin B12**

**H. Vitamin K**

# CORRECT ANSWERS

1. H
2. B
3. A
4. C
5. E

Vitamin K is required for formation of the γ-carboxyglutamate residues in several proteins required for blood clotting. Consequently, a deficiency of vitamin K results in a tendency to bleed. Niacin deficiency is characterized by the three Ds: diarrhea, dermatitis, and dementia (and death if untreated). Folic acid deficiency can result in neural tube defects in the developing fetus. Night blindness is one of the first signs of vitamin A deficiency. Rod cells in the retina detect white and black images and work best in low light, for example, at night. Rhodopsin, the visual pigment of the rod cells, consists of 11-cis retinal bound to the protein opsin. Vitamin C is required for the hydroxylation of proline and lysine during collagen synthesis. Severe vitamin C deficiency (scurvy) results in defective connective tissue, characterized by sore and spongy gums, loose teeth, capillary fragility, anemia, and fatigue.

2. A 52-year-old woman presents with fatigue of several months' duration. Blood studies reveal a macrocytic anemia, reduced levels of hemoglobin, elevated levels of homocysteine, and normal levels of methylmalonic acid. Which of the following is most likely deficient in this woman?

A. Folic acid

B. Folic acid and vitamin B12

C. Iron

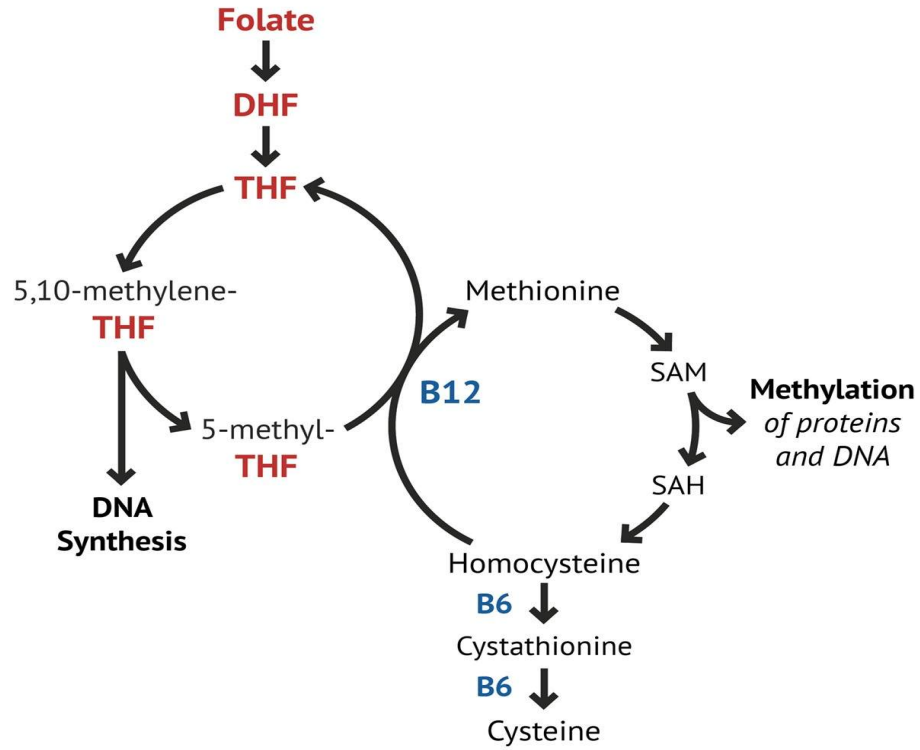
D. Vitamin C

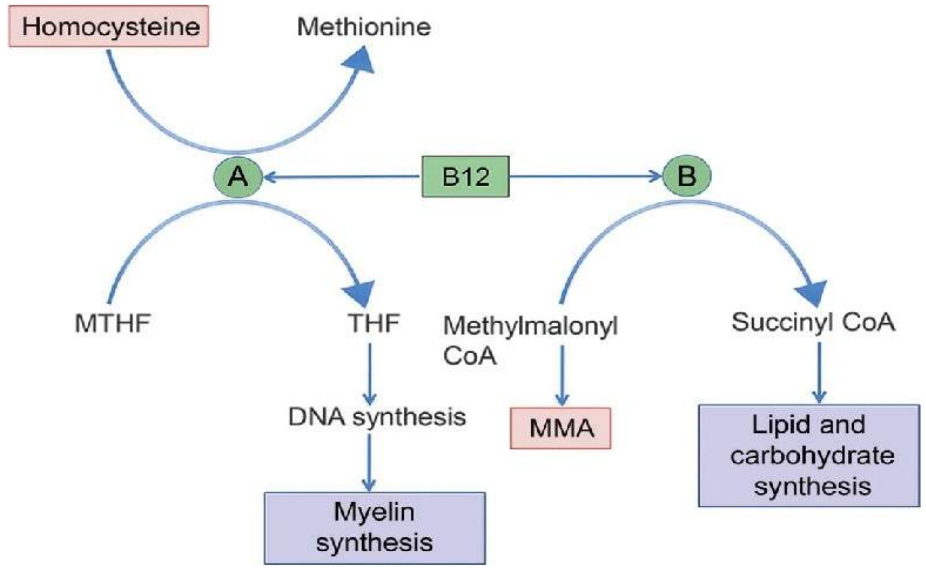
**Correct answer = A**

Macrocytic anemia is seen with deficiencies of folic acid, vitamin B12, or both.

Vitamin B12 is utilized in only two reactions in the body: the remethylation of homocysteine to methionine, which also requires folic acid (as tetrahydrofolate [THF]), and the conversion of methymalonyl coenzyme A to succinyl coenzyme A, which does not require THF.

The elevated homocysteine and normal methylmalonic acid levels in the patient's blood reflect a deficiency of folic acid as the cause of the macrocytic anemia.



3. A 10-month-old African-American girl, whose family recently located from Maine to Virginia, is being evaluated for the bowed appearance of her legs. The parents report that the baby is still being breastfed and takes no supplements. Radiologic studies confirm the suspicion of vitamin D–deficient rickets. Which one of the following statements concerning vitamin D is correct?

A. A deficiency results in an increased secretion of calbindin.

B. Chronic kidney disease results in overproduction of 1,25-dihydroxycholecalciferol

(calcitriol).

C. 25-Hydroxycholecalciferol (calcidiol) is the active form of the vitamin.

D. It is required in the diet of individuals with limited exposure to sunlight.

E. Its actions are mediated through binding to G protein–coupled receptors.

F. It opposes the effect of parathyroid hormone.

**Correct answer = D.**

Vitamin D is required in the diet of individuals with limited exposure to sunlight, such as those living at northern latitudes like Maine and those with dark skin.

Note that breast milk is low in vitamin D, and the lack of supplementation increases the risk of a deficiency.

Vitamin D deficiency results in decreased synthesis of calbindin.

Chronic kidney disease decreases production of calcitriol (1,25-dihydroxycholecalciferol), the active form of the vitamin.

Vitamin D binds to nuclear receptors and alters gene transcription. Its effects are synergistic with parathyroid hormone.

4. A newborn baby is typically given a vitamin K injection immediately after birth because colonization of the intestine with normal flora organisms that normally produce this vitamin not yet adequate. Unfortunately, supplementation of vitamin K to the pregnant mother does not affect the vitamin level in the fetus. Which of the following adverse effects may occur if the child does not receive the vitamin K injection?

a. Alopecia
b. Cerebral hemorrhage
c. Dermatitis
d. Night blindness
e. Rickets

**Correct answer = B**

Newborns are given a vitamin K injection at birth to prevent the rare occurrence of hemorrhagic disease of the newborn. Unlike other lipid-soluble vitamins, vitamin K requires a well – developed gut flora to be synthesized. Vitamin K is required for the coagulation cascade. In older individuals, vitamin K deficiency can lead to easy bruising, gum bleeding and an increased prothrombin time.

Alopecia results from biotin deficiency, which typically occurs from excessive consumption of raw eggs containing the biotin-binding protein.

Dermatitis along with dementia and diarrhea are components of pellagra which is caused by niacin deficiency.

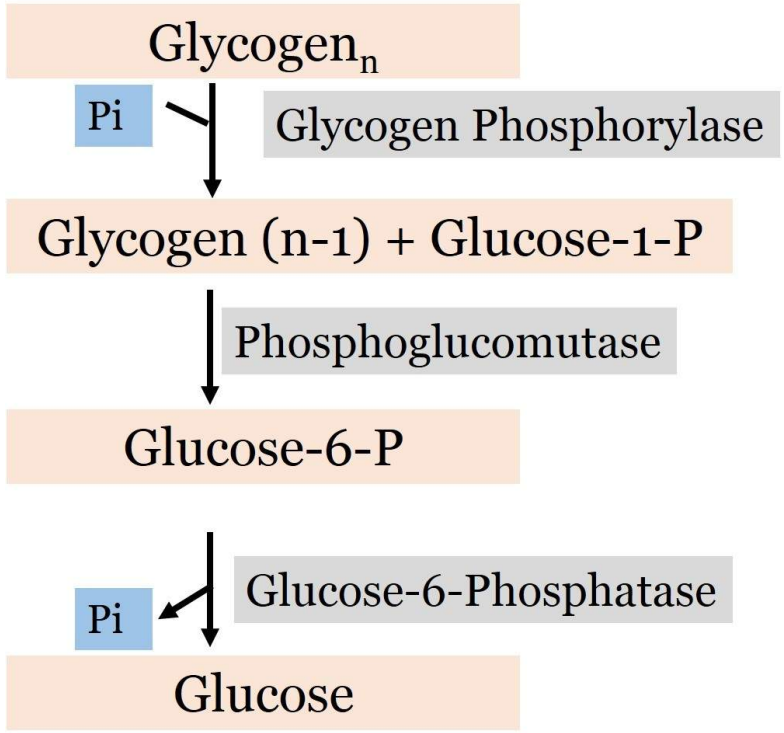
Night blindness can result from vitamin A deficiency.

Rickets is often caused by vitamin D deficiency, leading to poor calcium absorption.

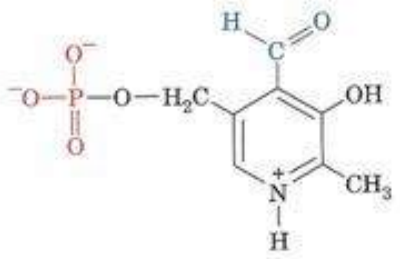
5. Why might a deficiency of vitamin B6 result in a fasting hypoglycemia? Deficiency of what other vitamin could also result in hypoglycemia?

Vitamin B6 is required for glycogen degradation by glycogen phosphorylase. A deficiency would result in fasting hypoglycemia.

Additionally, a deficiency of biotin (required by pyruvate carboxylase of gluconeogenesis) would also result in fasting hypoglycemia.




**Phosphorylase has a "random sequential"enzyme mechanism that involves PLP (pyridoxyl-5'-phosphate), a vitamin B6 derivative:**



Pyridoxal-5'-phosphate (PLP)

# THANK YOU